GSCF Data Release April 2016 (through 2015 wave)

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Abstract

The Gansu Survey of Children and Families (GSCF) is a longitudinal, multi-level study of rural children, including education, health, psycho-social development, and adult outcomes. Data have been collected in 2000, 2004, 2007, 2009, and 2015. This document introduces the available data files across all waves and discusses how to link data from the different waves together.

1 GSCF Survey Questionnaires

The GSCF data is comprised of the following surveys:

- Child surveys: multi-stage cluster target sample of 2000 rural children in Gansu aged 9-12 in the year 2000 (+oldest younger school-aged sibling in 2004) (+ new sample of 1,400 children in same villages aged 9-15 in 2007)
- Child assessments and surveys of children's households, mothers, and fathers linked to the target sample in 2000 and 2004, sibling sample in 2004, and new sample in 2007
- Census of primary and middle school teachers & principals in the main primary and middle schools attended by children in each of the 100 sample villages
- Community surveys of village leaders (all waves except 2009) (+ township health clinic directors in 2004)

2 Data File Names

A full list of all available data files is described in Table 1.

In Table 1, each file name is comprised of 3 parts: questionnaire name + year of the survey + sample. We have three different child samples in the data:

- t - target kids from the original cohort 2000 (surveyed in 2000, 2004, 2009 and 2015)
- s - oldest younger siblings of target kids who were of school age (≥ 6) in 2004 (surveyed in 2004 and 2015)
- n - new sample of kids aged 9-15 in 2007 (surveyed in 2007)

The data files ending with "ts" include data for both target kids and siblings.

3 Define Unique Identifiers

For all waves, we include the following identifying codes in the raw data:

- a1 - county code (2 digits)
- a2 - township code (2 digits)
- a3 - village code (2 digits)
- a4 - family code (2 digits)
- a5 - school code (2 digits)
- a6 - teacher code (3 digits)

Based on the above codes, we defined household identifiers (hhid = $a1 \times 10^6 + a2 \times 10^4 + a3 \times 10^2 + a4$), school identifer (schid =$a1 \times 10^6 + a2 \times 10^4 + a3 \times 10^2 + a5$) and teacher identifier (teachid = $a1 \times 10^8 + a2 \times 10^6 + a3 \times 10^4 + a5 \times 10^2 + a6$). These identifiers can be used to link data within or across waves.

4 Linking Individual Data Using Personal Identifying Codes (pid$**$)

A special data file GSCF_merge_pid.dta[1] has been created to facilitate merging individual data on children, parents, and siblings across waves. The file contains information on all such individuals who appear in any of the survey waves, each individual in a separate row. It includes a personal identifier code for each survey year. To link individual data found in the household questionnaire data file, for each wave arrange the individual data in the household roster of the household questionnaire in long form, so that the data for each individual is in a separate row. You will need to create the personal identifier code (named pid$**$, where $**$ is the survey year) equal to the row number in the household roster (and indicated in the variable names for household roster data)[2].

For the convenience of users, we have also added the pid$**$ variables to all child-specific data files to facilitate linking such files to child-specific data from the household questionnaires. Individual-specific data from other tables (not the household roster) in the household questionnaires can be linked using the row number found in the first column of each table (which can be defined as pid$**$ for linking purposes). If you have two individual-level data files, for example ind2000.dta and ind2004.dta, then the simplest way to merge them is to merge each file with GSCF_merge_pid.dta, using pid00 to merge the data from ind2000.dta, and pid04 to merge the data from ind2004 as follows:

```
**sample code.**
use GSCF_merge_pid.dta
sort hhid pid00
merge hhid pid00 using ind2000.dta
sort hhid pid04
merge hhid pid04 using ind2004
```

Please note that the pid codes are year-specific, individuals do not necessarily have the same pid in different years. This is why you must use GSCF_merge_pid.dta to link individual data across waves. GSCF_merge_pid.dta also includes variables that may be useful for users to assess themselves the

[1] See Table 5 for the data codebook.

[2] Note: Using "reshape long" command in STATA to define "pid$**$" as the index variable. For example, "reshape long varname, i(hhid) j(pid00)".

quality of linking across data waves, which is discussed below. In addition, GSCF_merge_pid.dta also includes variables from** and hhfrom**, which indicate whether the child is in the household roster and whether a household is surveyed in a particular year, respectively. We also create an indicator "target_sib04" to identify if the child is in the sibling sample in 2004.

5 Same Households Surveyed?

We have analyzed the data across waves to diagnose the likelihood that children and households surveyed in later waves are actually the same children as those surveyed in the first wave. In the data file GSCF_merge_pid.dta, we define variables samehh** and samekid** (where ** is the survey year) to indicate the reliability that the household or child surveyed in year ** is the same as those surveyed in the first wave in 2000. The samehh** and samekid** variables take on the values 1, 2, or 3, where:

- 1 = "definitely the same household or child",
- 2 = "uncertain whether same household or child", and
- 3 = "probably not the same household or child".

To make these assessments, we examine the household demographics and family structure in different waves, including father's birth year, mother's birth year, as well as all children's gender, birth year and relationship code to the target children.

"samehh**" and "samekid**" equal one if two of the following three conditions are satisfied:

(a) both parents' birth year difference is ≤ 2 across waves;

(b) all kids' birth year difference is ≤ 2 across waves;

(c) the sibling structure is identical (i.e., all kids have consistently the same relation codes across waves)

"samehh**" and "samekid**" equal three if two of the following three conditions are satisfied:

(a) either father or mother's birth year difference is ≥ 4 across waves;

(b) at least one kid's birth year difference is ≥ 4 across waves;

(c) the sibling structure differs across waves

Finally, for all the remaining cases, specially in cases where "samehh$**$" and "samekid$**$" do not equal one or three are categorized as the "not-so-sure" group (i.e., we let "samehh$**$" and "samekid$**$" equal two).

6 Summary Statistics on Surveyed Households

Table 2 reports the number of households that can be successfully interviewed in later waves. Column (1) lists the number of households surveyed in each wave. Column (3) reports the number of the surveyed households which belong to the original 2000 sample; while column (2) reports the number of households which may not be in the original cohort 2000. Columns (4) - (10) present the number of households surveyed in later waves, conditional on their being surveyed in different combinations of previous waves. For example, we can see from column (4) that 1762 households were surveyed in 2000, 2004 and 2007; 1813 households were surveyed in 2000, 2004 and 2009; 1657 households were surveyed in 2000, 2004 and 2015, etc. Similarly, we can see that 1490 households were surveyed in all five waves (2000, 2004, 2007, 2009 and 2015) from column (10).

Table 3 reports the number of kids (sample children and siblings) and households defined as the same ones across all available waves. In the first row of table 3, we can see that among the kids in the household roster across all 5 waves, 3470 are defined as identical kids across all years; 306 of them are probably not the same across years; 36 are unclear cases. If we focus on the 2000 target kids sampled in 2000, we can see from column (4)-(6) that 1490 households of target kids were surveyed in all five waves. Among them, 1383 are verified as the same one as in 2000; 97 likely differ across waves; and 10 are in the "not-sure" category. Similarly, we can see the number of kids and households (only surveyed in 4, or 3, or 2 waves) categorized into the 3 groups.

Table 4 reports the number of target kids and target siblings which can be traced in later waves. We can see from row (1) that 1918 target kids in the original Wave 2000 were also surveyed in 2004; however, only 1766 of

the 2000 target kids were sampled in 2007 and this number becomes even smaller in the last cohort, when we have only 1613 children from the 2000 target kids sampled in 2015. Row (2) -(4) reports how many of these target kids can be categorized into three groups: same, different, and not-so-sure. In panel 2 of table 4, we report the number of target siblings (first sampled in 2004) listed in the household roster in later waves. We also report how many of them identified as the same children across waves in row (6). Here we can see that around 80% of target siblings were surveyed again in 2015.

7 More Information

The following link gives more information about the GSCF data. `https://china.pop.upenn.edu/`

Table 1. GSCF Survey Questionnaires and Data Files

		2000	2004	2007	2009	2015
Child	Child Questionnaire	child2000t.dta	child2004t.dta child2004s.dta	child2007n.dta	child2009t.dta	child2015t.dta child2015s.dta
Child Assessments	Math/Lang.	achieve2000t.dta	achieve2004t.dta; achieve2004s.dta			
	Life-skills/Literacy	literacy2000t.dta	literacy2004t.dta	literacy2007n.dta	literacy2009t.dta	
	Health		health2004t.dta; health2004s.dta			
	Homeroom teacher	hrtea2000t.dta	hrtea2004t.dta; hrtea2004s.dta	hrtea2007n.dta		
Child's Family	Household	hh2000ts.dta	hh2004ts.dta	hh2007ts.dta hh2007n.dta		hh2015ts.dta
	Mother	moth2000ts.dta	moth2004ts.dta	moth2007n.dta		
	Father		fath2004ts.dta	fath2007n.dta		
Schools & Teachers	Teacher	teach2000.dta	teach2004.dta	teach2007.dta		
	School Principals	prin2000.dta	prin2004.dta	prin2007.dta		
	School Finance			schfin2007.dta		
Local Contexts	Village	vill2000.dta	vill2004.dta	vill2007.dta		vill2015.dta
	Township Clinic		clinic2004.dta			
	County Ed. Bureau			county2007.dta		

Notes: The file names with suffix "t" are for target kids surveyed in the original cohort 2000. Suffix "s" is for the target siblings first surveyed in 2004. Suffix "n" is the data set for kids in the 2007 new cohorts.

Table 2. Number of Households Surveyed across Waves of GSCF

Wave X=	total number of households	households not in the original 2000 sample	conditional on households surveyed in previous waves							
			2000	2000-2004	2000-2007	2000-2004-2007	2000-2009	2000-2004-2009	2000-2007-2009	2000-2004-2007-2009
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
2000	2000	0	2000	*	*	*	*	*	*	*
2004	1918	0	1918	*	*	*	*	*	*	*
2007	1779	1	1778	1762	*	*	*	*	*	*
2009	1862	8	1854	1813	1681	1670	*	*	*	*
2015	1695	0	1695	1657	1548	1536	1634	1604	1500	1490

Notes: This table reports how many of the households surveyed in Wave X also appears in other earlier waves. Column (1) reports the number of households surveyed in each wave. Column (2) counts the number of households surveyed in Wave X but not in the original Cohort 2000. Column (3) reports the number of households surveyed in both Wave 2000 and Wave X. Column (4) counts the number of households surveyd in 2000, 2004 and Wave X. Column (5) lists the number of households surveyed in Wave 2000, 2007 and Wave X. Column (6) counts the number of households surveyed in 2000, 2004, 2007 and Wave X. Column (7) reports the number of households surveyed in Wave 2000, 2009 and X. Column (8) reports the number of households surveyd in Wave 2000, 2004, 2009 and X. Column (9) reports the number of households surveyed in 2000, 2007, 2009 and Wave X. Column (10) reports the number of households surveyed in Wave 2000, 2004, 2007, 2009 and X. To avoid repetition, we only report the data in the lower diagonal matrix and use * to denote the other cells.

Table 3. Number of kids and households verified as the same one across all available waves

Data Source		number of kids			among 2000 households of target kids		
		same	not sure	different	same	not sure	different
		(1)	(2)	(3)	(4)	(5)	(6)
data in all 5 waves	00,04,07,09,15	3,470	36	306	1,383	10	97
data in 4 waves	00,04,07,09	398	5	48	163	2	15
	00,04,07,15	89	6	17	36	3	7
	00,04,09,15	274	2	41	101	1	12
	00,07,09,15	25	0	0	10	0	0
data in 3 waves	00,04,07	100	0	3	45	0	1
	00,04,09	63	3	11	23	1	5
	00,04,15	17	0	0	7	0	0
	00,07,09	2	0	0	1	0	0
	00,07,15	3	0	2	1	0	1
	00,09,15	36	10	4	16	3	1
data in 2 waves	00,04	14	0	0	6	0	0
	00,07	6	0	0	3	0	0
	00,09	8	8	8	4	3	3
	00,15	13	0	4	5	0	1
TOTAL		4,518	70	444	1,804	23	143

Notes: This table reports the number of all kids and households verified as the same one across all available waves. Column (1) reports the number of households which are verified as the same household across all the waves the household surveyed. Column (2) reports the number of households which we are not sure if they are the same families being surveyed across the waves. Column (3) reports the number of families verified as the different households being surveyed across waves. Column (4) reports the number of households which are only being surveyed once and thus the definition "same household" is not well defined. Similarly, column (5)-(8) reports the number of kids verified as the same, not sure, different or undefined. However, column (5)-(8) focus on the 2000 sampled kids from the original Cohort 2000.

Table 4: Track Target Kids and Target Siblings across Waves

			Survey Data in the Year =			
			2004	2007	2009	2015
			Panel 1: Target Kids			
(1)	Number of the original Wave 2000 households surveyed		1918	1766	1801	1613
	Target kids sampled in 2000?					
(2)		-yes	1914	1742	1595	1445
(3)		-not sure	2	5	50	44
(4)		-no	2	19	156	124
			Panel 2: Target Siblings			
(5)	Target siblings sampled in 2004		840	742	746	705
	Same kids sampled in 2000?					
(6)		-yes	840	734	691	663
(7)		-not sure	0	2	9	11
(8)		-no	0	6	46	31

Notes: Our sample of target siblings is restricted to kids in both the child survey and the household survey in the year 2004. Row (1) reports the number of target kids in each wave whose families were also sampled in the original Wave 2000. Row (2) reports the number of the target kids in each wave, who are verified as the same kids in the original Wave 2000. Row (3) reports the number of target kids in each wave, whose families are also sampled in the original Wave 2000, but we are not sure whether we get the same family or the same target kid. Row (4) reports the number of target kids in each wave, whose familes are also sampled in Wave 2000 (becasue the household identifiers are the same), but they should be different families according to our definition of "same household". Row (5) reports the number of target siblings (i.e., the siblings who filled the child survey in the year 2004) showing up in the household survey in each wave. Row (6) reports the number of target siblings apearing in each wave's household survey who are verified to be the same kids as we find in the 2004 child survey.

Table 5. Codebook of Data GSCF_merge_pid.dta

variable name	notes
hhid	household identifier
pid00	individual identifier in 2000
pid04	individual identifier in 2004
pid07	individual identifier in 2007
pid09	individual identifier in 2009
pid15	individual identifier in 2015
from00	=1, if the kid is in household roster in 2000
from04	=1, if the kid is in household roster in 2004
from07	=1, if the kid is in household roster in 2007
from09	=1, if the kid is in household roster in 2009
from15	=1, if the kid is in household roster in 2015
hhfrom00	=1, if the household is surveyed in 2000
hhfrom04	=1, if the household is surveyed in 2004
hhfrom07	=1, if the household is surveyed in 2007
hhfrom09	=1, if the household is surveyed in 2009
hhfrom15	=1, if the household is surveyed in 2015
target_sib04	=1, if the kid is the oldest younger siblings of target kids who where of school age in 2004
relat00	relationship code in 2000
byear00	birth year in 2000
relat04	relationship code in 2004
byear04	birth year in 2004
relat07	relationship code in 2007
byear07	birth year in 2007
relat09	relationship code in 2009
byear09	birth year in 2009
relat15	relationship code in 2015
byear15	birth year in 2015
fbyear00	father's birth year in 2000
mbyear00	mother's birth year in 2000
fbyear04	father's birth year in 2004
mbyear04	mother's birth year in 2004
fbyear07	father's birth year in 2007
mbyear07	mother's birth year in 2007
fbyear09	father's birth year in 2009
mbyear09	mother's birth year in 2009
fbyear15	father's birth year in 2015
mbyear15	mother's birth year in 2015
samehh0004	indicator to define if the household is same or not between wave 2000 and 2004
samehh0007	indicator to define if the household is same or not between wave 2000 and 2007
samehh0009	indicator to define if the household is same or not between wave 2000 and 2009
samehh0015	indicator to define if the household is same or not between wave 2000 and 2015
samekid0004	indicator to define if the individual is same or not between wave 2000 and 2004
samekid0007	indicator to define if the individual is same or not between wave 2000 and 2007
samekid0009	indicator to define if the individual is same or not between wave 2000 and 2009
samekid0015	indicator to define if the individual is same or not between wave 2000 and 2015

Notes: This data file is created to facilitate merging individual data across waves.